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Sta Probability & Stochastic Processes - Rene Wang Siwei (0081191)

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1. a) Given r.v $X \sim N(5, 2^2)$ & $Y = 2X + 4$

To find μ_Y, σ_Y & f_Y :

$$F_Y(y) = P(Y \le y) = P(2X+4 \le y) = P\left(X \le \frac{y-4}{2}\right) = F_X\left(\frac{y-4}{2}\right)$$

$$\therefore f_Y(y) = \frac{1}{2} f_X\left(\frac{y-4}{2}\right) = \frac{1}{\sqrt{2\pi}4} e^{-\frac{\left(\frac{y-4}{2}-5\right)^2}{2(4)}}$$

$$= \frac{1}{\sqrt{2\pi(16)}} e^{\frac{(y-14)^2}{2(16)}}$$

Since for Gaussian distribution, $f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}$: ✓

$\mu_Y = 14$, $\sigma_Y = 4$ ✓

b) Given r.v X is uniform in $(0,1)$ & $Y = -\ln X$

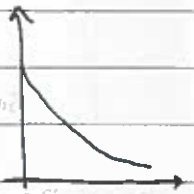
$$F_X(x) = \begin{cases} \frac{x}{1-0} = x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$Y = -\ln X$

$$F_Y(y) = P(Y \le y) = P(-\ln X \le y) = P(X \ge e^{-y})$$

$$= 1 - P(X \le e^{-y})$$

$$= 1 - F_X(e^{-y})$$



$$\therefore F_Y(y) = 1 - e^{-y}$$

$$f_Y(y) = \frac{d F_Y(y)}{dy}$$

$$= \begin{cases} e^{y}, & 0 < y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

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c) Given r.v X & Y are iid with:

$$f_X(x) = e^{-x}, \quad x>0$$

$$f_Y(y) = e^{-y}, \quad y>0$$

To Find $f_Z(z)$:

i) $Z = X+Y$

~~Since X & Y are i.i.d r.vs~~

~~$f_{X,Y}(x,y) = e^{-x}e^{-y} f_X(x) \cdot f_Y(y)$~~

~~$= e^{-(x+y)}$~~

Note that $f_Z(z)$ is the convolution of $f_X(x)$ & $f_Y(y)$:

$$f_Z(z) = \int_0^z f_X(z-y) f_Y(y)\, dy$$

$$= \int_0^z e^{-(z-y)} e^{-y}\, dy$$

$$= \int_0^z e^{-z}\, dy$$

$$= ze^{-z} \quad \text{for } z>0$$ ✓

ii) $Z = \min(X,Y)$

$$F_X(x) = \int_0^x e^{-x}\, dx$$

$$= [-e^{-x}]_0^x$$

$$= 1-e^{-x} \text{ for } x>0$$

Similarly, $F_Y(y) = 1-e^{-y}$ for $y>0$.

$$F_Z(z) = P(Z \le z) = 1 - P(Z > z)$$

$$= 1 - P(\min(X,Y) > z)$$

$$= 1 - P(X \ge z)\, P(Y > z)$$

$$= 1 - (1-P(X \le z))(1-P(Y \le z))$$

$$= 1 - e^{-(z+z)}$$

$$= 1 - e^{-2z}$$

$$f_Z(z) = 2e^{-2z} \quad \text{for } z>0$$ ✓

iii) $Z = \max(X,Y)$

$$F_Z(z) = P(Z \le z) = P(\max(X,Y) \le z)$$

$$= P(X \le z)\, P(Y \le z)$$

$$= (1-e^{-z})(1-e^{-z}) = 1-2e^{-z}+e^{-2z}$$

$$f_Z(z) = 2e^{-z} - 2e^{-2z} \quad \text{for } z>0.$$ ✓

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2. a) Given r.v X has $f_{X_i}(x) \sim c^4 x^3 e^{-cx}$, $x>0$ { i.i.d samples $x_i = 3.1, 3.4, 3.3$.

To find maximum-likelihood estimate of c:

For n samples, we get

$$f(x;c) = \sum_{i=0}^{n} c^4 x_i^3 e^{-cx_i}$$

$$= c^{4n} (x_1 \dots x_n)^{3n} e^{-c(x_1+\dots+x_n)}$$

$$\frac{\partial f(x;c)}{\partial c} = 4nc^{4n-1} (x \dots x_n)^{3n} e^{-c(x_1+\dots+x_n)} - (x_1+\dots+x_n) c^{4n} (x_1 \cdots x_n)^{3n} e^{-c(x_1+\dots+x_n)}$$

$$= \left(\frac{4n}{c} - (x_1+\dots+x_n)\right) c^{4n} (x_1 \dots x_n)^{3n} e^{-c(x_1+\dots+x_n)}$$

$$= \left(\frac{4n}{c} - (x_1+\dots+x_n)\right) f(x;c)$$

To get maximum, $\frac{\partial f(x;c)}{\partial c} = 0$

$$\frac{4n}{c} = x_1 + \dots + x_n$$

Given $n = 3$, $x_i = 3.1, 3.4, 3.3$, : $\frac{12}{c} = 3.1 + 3.4 + 3.3$

$$c = \frac{12}{9.8} = 1.2245$$

b) Given X_i are i.i.d and uniform on $(0,1)$ & $Y = \max X_i$

$$f_{X_i}(x) = \begin{cases} \frac{1}{1-0} = 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$F_{X_i}(x) = \begin{cases} x, & 0 \le x \le 1 \\ 0, & x < 0 \end{cases}$$

To find $F(y)$, $F_Y(y) = P(\max(X_1, X_2 \dots, X_n) \le y)$

$$= P(X_1 \le y, X_2 \le y, \dots, X_n \le y) = \prod_{i=1}^{n} P(X_i \le y)$$

$$= \prod_{i=1}^{n} F_{X_i}(y)$$

$$= y^n, \quad 0 \le y \le 1 \quad \text{(shown)}$$

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